

4. Graph Theory :

①

When ckts are more elaborate, KCL, KVL & V-I relationship is not enough. we can use mesh & nodal analysis then

Elements of graph theory :

1) Branch & Node

branch which represents a ckt element, is a line joining two nodes.

* a node is a common point where 2 or more branches meet.

2) Graph.

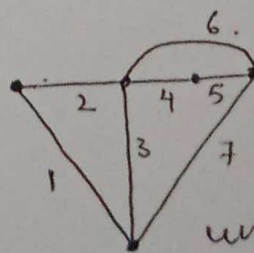
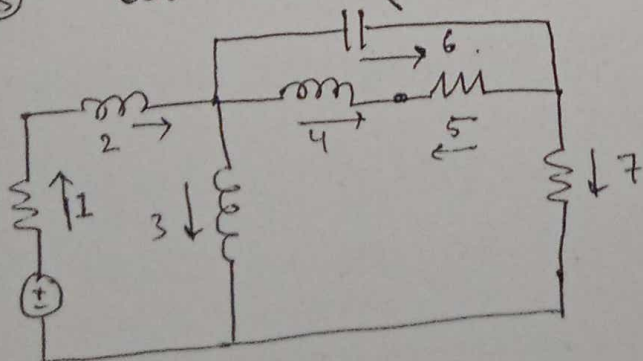
linear graph → collection of nodes & branches, & shows for interconnection of the element of a N/w.

~~con~~ graph is connected iff. there is a path directly or indirectly b/w every pair of node.

Properties → ① max. no of branches = no. of nodes.

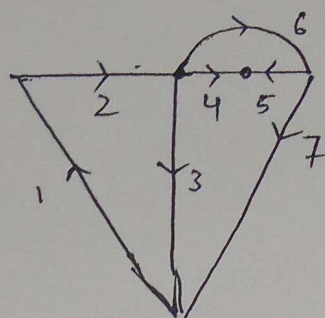
② exactly 2 paths b/w any pair of node in a ckt.

③ at least 2 branches in a ckt.



undirected graph.

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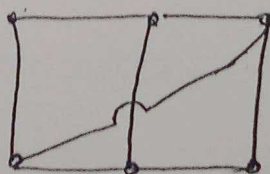
oriented/ directed graph.

Rank of the graph $= n-1$

where $n \Rightarrow$ no. of nodes.

Planar graph $\rightarrow$ is drawn by keeping all pts of intersection of 2 or more branches as nodes & representing the n/w elements by lines, ideal voltage source & ideal I source are replaced by s.c & o.c resp.

Non-planar graphs $\rightarrow$



3) Path $\therefore \rightarrow$

An ordered seq. of branches traversing from one node to another node.

④ Tree (twigs) & Co-tree (links or chords). (2)

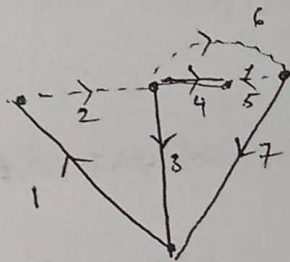
Tree $\rightarrow$ Connected graph with no closed path.
or any set of branches in the original graph that is just sufficient to connect all nodes.

$\rightarrow$ branches in a tree $= n-1 = \text{twigs} = n_t$

$\rightarrow$ remaining branches = links $= b - n_t = b - n_t + 1 = n_l$

$\rightarrow$ links form a ^{sub-}graph $\rightarrow$ Co-tree i.e. Complement of the tree

graph $\rightarrow$ tree + Co-tree.



Properties of tree

① In a tree, there ~~is~~ exists only one path b/w any pair of nodes.

② Every connected graph has at least 1 tree.

③ Rank of tree $= n-1 = \text{rank of graph}$.

⑤ Loop or circuit:

If a link is added to a tree, the resulting graph contains one closed path called loop.

Properties of loop

① There exists at least 2 branches in a loop.

② max ^{possible} branches in a loop = no. of nodes in graph.

Fundamental-loop :: (f-loop) or tie sets or basic loops ::
→ loops which contain only 1 link are independent

→ no. of f-loops = no. of links

→ orientation of f-loop is same as its link.

⑥ Cut-set ::

minimal set of branches that if removed divides a connected graph into two connected sub graphs.

→ separates the nodes of the graph into 2 groups.

$$\text{rank} = (n-1) - 1$$

f-cutsets / basis or fundamental cutsets ::

→ Cutsets containing only 1 twig are independent.

→ orientation of a f-cutset is chosen to be same as that of its twig.

→ no. of f-cutsets = no. of twigs.

(3)

INCIDENCE MATRIX [A]:

for a graph with 'n' nodes & 'b' branches

Complete incidence matrix $A_c(n \times b)$

$A_{nb} = +1$; b leaves node n

$= -1$; b enters node n

$= 0$; b not incident with n.

$$A_c = \begin{array}{c|ccccccc} \text{Nodes} & \text{branches} \rightarrow \\ \downarrow & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \begin{array}{c} a \\ b \\ c \\ d \\ e \end{array} & \left[\begin{array}{ccccccc} -1 & +1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & +1 & +1 & 0 & +1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & +1 & -1 & +1 \\ +1 & 0 & -1 & 0 & 0 & 0 & -1 \end{array} \right] \end{array}$$

→ Since each branch enters a single node & leaves a single other node, so each column contains a single -1 & +1 with all other elements = 0.

By eliminating any row, we get the reduced [A]. Dimension $= (n-1) \times b = n_t \times b$.

rank = $n-1$ or n_t .

Reduced [A]

$$\begin{array}{c|ccccccc} N & B \rightarrow \\ \downarrow & 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ \begin{array}{c} a \\ b \\ c \\ d \end{array} & \left[\begin{array}{ccccccc} -1 & +1 & 0 & 0 & 0 & 0 & 0 \\ 0 & -1 & +1 & +1 & 0 & +1 & 0 \\ 0 & 0 & 0 & -1 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & +1 & -1 & +1 \end{array} \right] \end{array}$$

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$$A = \begin{matrix} & \begin{matrix} N \\ \downarrow \end{matrix} & \begin{matrix} B_1 & \vec{3} & 4 & 7 & 1 & 2 & 5 & 6 \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} -1 & 0 & 0 & 0 & +1 & 0 & 0 \\ 0 & +1 & +1 & 0 & -1 & 0 & +1 \\ 0 & 0 & -1 & 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & +1 & 0 & +1 & -1 \end{bmatrix} \end{matrix}$$

or $A = [A_t : A_l]$

$A_t \rightarrow$ sp. matrix of dim $(n-1) \times (n-1) \rightarrow$ its cl $\rightarrow$ twigs

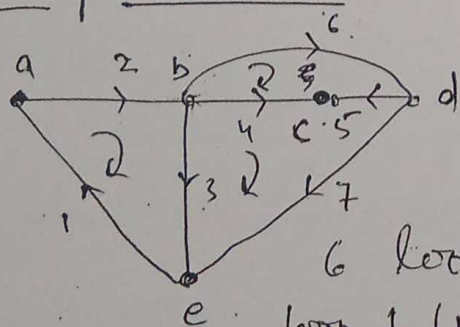
$A_l \rightarrow (n-1) \times (b-n+1) = nt \times nl$
its clmn $\rightarrow$ links.

Since $|A_t| = 1$ & rank of $A_t = n-1$ or nt .

Properties:

- ① sum of entries in any column = 0.
- ② rank of $A_c = n-1$
- ③ $\Delta |A_c| = 0$.

Loop matrix or Circuit-Matrix $[B]$.



$b_l b = 1$ if b is in loop l & their orientation coincide
 -1 if b is in loop l but don't coincide
 $= 0$ no in loop l

loop 1 (1, 2, 3)

loop 2 (4, 6, 5)

loop 3 (3, 4, 5, 7)

loop 4 (1, 2, 6, 7)

loop 5 (3, 6, 7)

loop 6 (1, 2, 4, 5, 7)

$$B_c = \begin{matrix} \begin{matrix} l \\ \downarrow \end{matrix} & \begin{matrix} \text{Branches} \\ \rightarrow \end{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Here, rows are not ^{linearly} independent.

$$\text{rank of } B_c = nl = b - n + 1.$$

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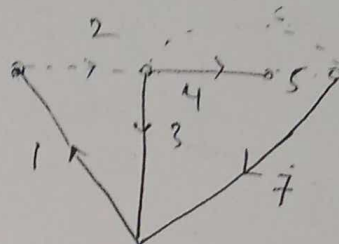
* $\rightarrow$ these nl loops are the minimal set required to describe the graph.

Tiesets

Tieset 2: $[1, 2, 3]$:

Tieset 5: $[3, 4, 5, 7]$:

Tieset 6: $[3, 6, 7]$:



Tieset matrix

$$B_f = \begin{matrix} & \begin{matrix} \text{Tieset} \backslash \text{Branches} \\ 2 \quad 5 \quad 6 \end{matrix} \\ \begin{matrix} 1 & 3 & 4 & 7 & 1 & 2 & 5 & 6 \end{matrix} \\ \begin{bmatrix} 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & -1 & -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$$\text{rank} = nl = b - n + 1$$

$$[B_f] = [B_{ft} : B_{fl}] = [B_{ft} : U]$$

$[B_{fl}]$ = for links of size $(nl \times nl)$

= Identity matrix = U .

$[B_{ft}]$ = for twigs.

Cut-Set Matrix [Q]

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Cut Set Matrix $[Q_c]$:

elements are

$q_{cb} = 1$ if b is in cut set c & orientation coincides.
 $q_{cb} = -1$ if b is in cut set c & orientation doesn't coincide.
 0 not in cut set c .

for the given graph, 10 cutsets are possible & as follows:-

Cutset 1: (1, 2)	6 (2, 3, 5, 6)
" 2: (2, 3, 4, 6)	7 (1, 3, 4, 6)
3: (4, 5)	8 (4, 6, 7)
4: (5, 6, 7)	9 (1, 3, 5, 6)
5: (1, 3, 7)	10 (2, 3, 7)

$Q_c =$

Cutset / br $\rightarrow$	1	2	3	4	5	6	7
1	-1	1	0	0	0	0	0
2	0	-1	1	1	0	1	0
3							
4							
5							
6							
7							
8							
9							
10							

$Q_f =$

	1	2	3	4	5	6
136	1	0	1	0	0	1
256	0	1	0	0	1	-1
345	0	0	1	1	1	0

⑤

$$\text{rank} = nt = n-1.$$

⇒ we require only nt cutsets to describe a linear graph of n nodes.

f-cutsets: formed when only 1 twig is used.

→ no. of f-cutsets = $nt = n-1$

→ orientation of f-cutset is same as its twig.

cutset 1 : (1, 2)
 " 3 : (2, 3, 5, 6)
 " 4 : (4, 5)
 " 7 : (5, 6, 7)

$$Q_f = \begin{matrix} & \begin{matrix} C=5 & B=7 & 4 & 7 & 2 & 5 & 6 \end{matrix} \\ \begin{matrix} 1 \\ 3 \\ 4 \\ 7 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 1 & 0 & 0 & -1 & -1 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 & -1 \end{bmatrix} \end{matrix}$$

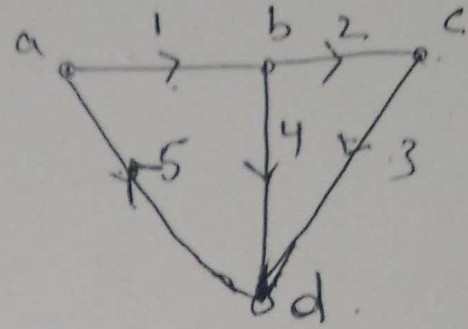
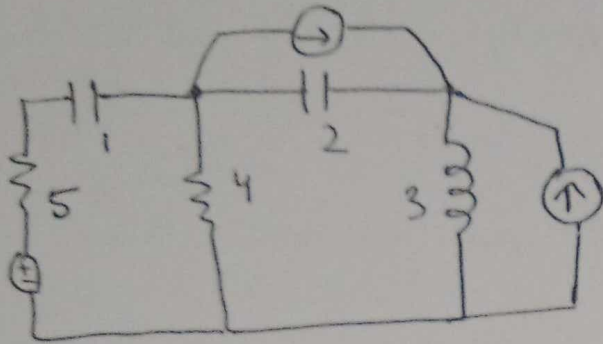
unit matrix

$$\text{rank} = nt = n-1.$$

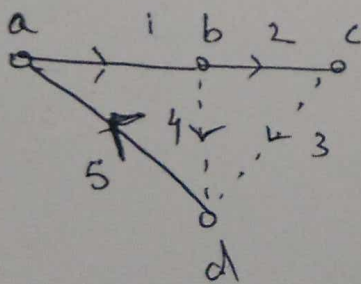
$$Q_f = [Q_{ft} : Q_{fl}] = [U : Q_{fl}]$$

⑤

Ques for the N/w shown. Determine all the tree & corresponding co-tree for the graph of the N/w. Then consider tree formed by br(1,2,5) using this tree write A, Bf & Qf.



S.No.	Tree (twigs)	Co-tree (links)
1	(1, 2, 5)	(3, 4)
2	(2, 4, 5)	(1, 3)
3	(3, 4, 5)	(1, 2)
4	(1, 2, 3)	(4, 5)
5	(1, 2, 4)	(3, 5)
6	(1, 3, 4)	(2, 5)
7	(1, 3, 5)	(2, 4)
8	(2, 3, 5)	(1, 4)



$$A_C = \begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline a & 1 & 0 & 0 & 0 & -1 \\ b & -1 & 1 & 0 & +1 & 0 \\ c & 0 & -1 & 1 & 0 & 0 \\ d & 0 & 0 & -1 & -1 & +1 \end{array}$$

$$A = \begin{array}{c|ccccc} & 1 & 2 & 3 & 4 & 5 \\ \hline a & 1 & 0 & 1 & 0 & 0 \\ b & -1 & 1 & 0 & 0 & 1 \\ c & 0 & -1 & 0 & 1 & 1 \\ d & 0 & 0 & 1 & 1 & 0 \end{array}$$

B_f (Tie-set matrix / f-loop matrix) (6)

$$\text{no. of Tie-sets} = \text{no. of link} = b - n + 1 = 5 - 4 + 1 = 2.$$

Tie-set 3 : (1, 2, 3, 5)

" 4 : (1, 4, 5)



$$B_f = \begin{array}{c} \text{Tie-set / br} \rightarrow \\ \begin{array}{c} 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cc|cc} 1 & 2 & 5 & 3 & 4 \\ \hline 1 & 1 & +1 & 1 & 0 \\ 1 & 0 & +1 & 0 & 1 \end{array} \end{array} \end{array} = [B_{ft} : U]$$

Q_f (f-cutset matrix)

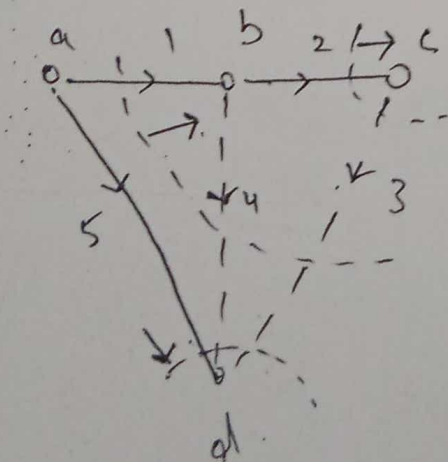
$$\text{No. of Cutsets} = \text{no. of twigs} = nt = n - 1 = 3.$$

Cutset 1 : (1, 3, 4) Cutset 5 : (3, 4, 5)

Cutset 2 : (2, 3)

Cutset / br $\rightarrow$

$$Q_f = \begin{array}{c} \begin{array}{c} 1 \\ 2 \\ 5 \end{array} \begin{array}{c} \begin{array}{cc|cc} 1 & 2 & 5 & 3 & 4 \\ \hline 0 & 0 & 0 & -1 & -1 \\ 0 & 1 & 0 & -1 & 0 \\ 0 & 0 & 1 & -1 & -1 \end{array} \end{array} \end{array} = [U : Q_{ft}]$$



Inter relation among various matrices :

The 3 diff. matrices are not independent.
As they are inter-related.

① Relation b/w A & B_f .

For a graph, 2 matrices A & B_f are
orthogonal to each other. i.e. if the columns of these
2 matrices are arranged in
the same order.
i.e. $A B_f^T = 0$.

$$\begin{bmatrix} A_t & A_\ell \end{bmatrix} \begin{bmatrix} B_{ft}^T \\ B_{f\ell}^T \\ 0 \end{bmatrix} = 0.$$

$$A_t B_{ft}^T + A_\ell B_{f\ell}^T = 0$$

$$B_{f\ell}^T = -A_t^{-1} A_\ell$$

$$\boxed{B_{f\ell} = [-A_t^{-1} A_\ell]^T}$$

② Relation b/w A & Q_f

$$A I_b = 0.$$

$$\& Q_f I_b = 0. = \textcircled{?} \left. \begin{array}{l} \text{follows} \\ \text{KCL} \end{array} \right\}$$

$$I_b = \begin{bmatrix} I_t \\ I_\ell \end{bmatrix} \text{ branch I of } b \times 1$$

$$\begin{bmatrix} A_t & A_\ell \end{bmatrix} \begin{bmatrix} I_t \\ I_\ell \end{bmatrix} = 0.$$

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$$A_t I_t + A_L I_L = 0$$

$$I_t = -A_t^{-1} A_L I_L \quad \text{--- (a)}$$

$$\star \quad [U : Q_{fL}] \begin{bmatrix} I_t \\ I_L \end{bmatrix} = 0.$$

$$I_t + Q_{fL} I_L = 0$$

$$I_t = -Q_{fL} I_L \quad \text{--- (b)}$$

$$Q_{fL} I_L = A_t^{-1} A_L I_L$$

$$\boxed{Q_{fL} = A_t^{-1} A_L}$$

③ b/w B_f & Q_f from ① & ②

$$B_f = [B_t : U] = [-Q_L^T U]$$

$$Q_f = [U : Q_L] = [U - B_t^T]$$

$$Q_{fL} = -B_{ft}^T \quad \text{or} \quad B_{ft} Q_f^T = [-Q_L^T U] \begin{bmatrix} U \\ Q_L^T \end{bmatrix} = 0$$

$$B_{ft} = -Q_{fL}^T \quad \text{--- (c)} \quad Q_f B_f^T = \underline{\underline{0}}$$

Relationships among parameters:

$$\text{KVL} : B_f V_b = 0$$

$$\text{KCL} : A I_b = 0$$

$$Q_f I_b = 0$$

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where $V_b = \begin{bmatrix} V_t \\ \vdots \\ V_l \end{bmatrix}$ & $I_b = \begin{bmatrix} I_t \\ \vdots \\ I_l \end{bmatrix}$

① b/w ~~cutset~~ V_l & V_t .

$$[B_{ft} : U] \begin{bmatrix} V_t \\ \vdots \\ V_l \end{bmatrix} = 0$$

$$B_{ft} V_t + V_l = 0$$

$$\text{or } \boxed{V_l = -B_{ft} V_t}$$

$$\text{or } \boxed{V_l = Q_{fL}^T V_t}$$

② b/w V_b & V_t .

$$V_b = \begin{bmatrix} V_t \\ \vdots \\ V_l \end{bmatrix} = \begin{bmatrix} U V_t \\ \vdots \\ Q_{fL}^T V_t \end{bmatrix} \quad \text{as } \begin{bmatrix} U \\ \vdots \\ Q_{fL} \end{bmatrix} = Q_f$$

$$\boxed{V_b = Q_f^T V_t}$$

③ b/w V_b & V_n (node voltage).

$$V_b = A^T V_n$$

as $V_b = V_i' - V_j'$

$\downarrow$ $\downarrow$
 volt at volt at
 node i node j

④ b/w I_t & I_L form (8)

$$I_t = -A_t^{-1} A_L I_L = -Q_{fL} I_L = B_{ft}^T I_L \quad \text{--- (C)}$$

⑤ b/w I_b & I_L

$$\boxed{I_b = B_f^T I_L}$$

$$I_b = \begin{bmatrix} I_t \\ I_L \end{bmatrix} = \begin{bmatrix} B_{ft}^T \\ 0 \end{bmatrix} I_L$$

$$= \underline{\underline{B_f^T I_L}}$$

Network Analysis

for a gen. n/w with b branches, V_b & I_b can be written as.

$$V_b = V_s + Z_b (I_b - I_s)$$

$$I_b = I_s + Y_b (V_b - V_s)$$

$V_b \rightarrow$ br. voltage mat. of $b \times 1$

$I_b \rightarrow$ br. I mat. of $b \times 1$

$V_s \rightarrow$ volt. source mat. of $b \times 1$

$I_s \rightarrow$ I. source mat. of $b \times 1$

$Z_b \rightarrow$ branch. imp. mat. of $b \times b$

$Y_b \rightarrow$ br. admittance mat. of $b \times b$

Loop Analysis:

br. volt. $\rightarrow$

$$V_b = V_s + Z_b (I_b - I_s)$$

x by B_f .

$$B_f V_b = B_f V_s + B_f Z_b (I_b - I_s)$$

$$0 = B_f V_s + B_f Z_b (B_f^T I_L - I_s)$$

$$B_f Z_b I_s - B_f V_s = B_f Z_b B_f^T I_L$$

$$\text{Let } B_f Z_b I_s - B_f V_s = V_L$$

$$B_f Z_b B_f^T = Z_L$$

$$V_L = Z_L I_L$$

matrix loop equation

Nodal Analysis:

$$I_n = Y_n V_n$$

$$I_n = A Y_b V_s - A I_s$$

$$Y_n = A Y_b A^T$$

matrix node eqⁿ

$$I_b = I_s + Y_b (V_b - V_s)$$

$$A I_b = 0 = A I_s + A Y_b (V_b - V_s)$$

$$V_b = A^T V_n$$

$$A Y_b V_s - A I_s = A Y_b A^T V_n$$

$$A Y_b V_s - A I_s = I_n$$

$$A Y_b A^T = Y_n$$

$$I_n = Y_n V_n$$

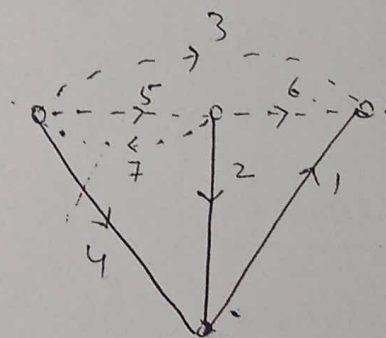
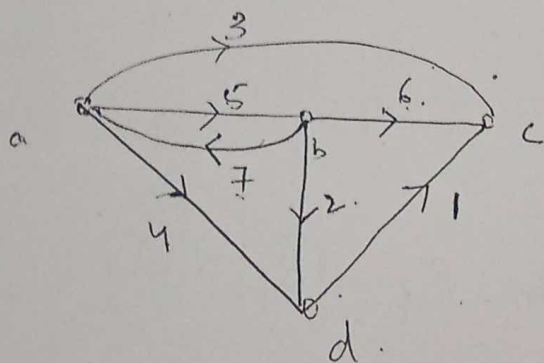
$$V_n = Y_n^{-1} I_n$$

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Ques . The reduced incidence matrix of a graph is given. Draw the graph & obtain f-loop & f-cutset matrices. verify result mathematically

$$A = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} a \\ b \\ c \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 & 1 & 1 \\ -1 & 0 & -1 & 0 & 0 & -1 & 0 \end{bmatrix} \end{matrix}$$

$$A_c = \begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \end{matrix} & \begin{bmatrix} 0 & 0 & 1 & 1 & 1 & 0 & -1 \\ 0 & 1 & 0 & 0 & -1 & 1 & 1 \\ -1 & 0 & -1 & 0 & 0 & -1 & 0 \\ 1 & -1 & 0 & -1 & 0 & 0 & 0 \end{bmatrix} \end{matrix}$$



f-loop no. of f-loops = no. of links = 4.

f-loop 3 (3, 1, 4) f-loop 6 (6, 1, 2)

f-loop 5 (5, 2, 4) f-loop 7 (2, 4, 7)

$$B_f = \begin{matrix} & \begin{matrix} \text{f-loop} \downarrow \end{matrix} & \begin{matrix} \text{br} \rightarrow & \text{twigs} & \leftarrow & \text{links} \rightarrow \end{matrix} \\ \begin{matrix} 3 \\ 5 \\ 6 \\ 7 \end{matrix} & \begin{bmatrix} 1 & 2 & 4 & 3 & 5 & 6 & 7 \\ -1 & 0 & -1 & 1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

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f-cutset

no. of f-cutset = no. of twigs = 3.

f-cutset 1 (1, 3, 6)

f-cutset 2 (2, 5, 6, 7)

f-cutset 4 (3, 4, 5, 7)

$$Q_f = \begin{matrix} & \begin{matrix} \text{f-cutset} \\ \downarrow \end{matrix} & \begin{matrix} \text{br} \rightarrow \\ 1 & 2 & 4 \end{matrix} & \begin{matrix} 1 & 3 & 5 & 6 & 7 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 4 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 1 \\ 1 & 1 & 0 & -1 & 1 \end{bmatrix} \end{matrix}$$

Verification

$$B_{ft} = [-A_t^{-1} A_e]^T \quad \& \quad B_f = [B_{ft} : U]$$

$$\underline{A}^{-1} \quad A_t = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$\Delta = \det[A_t] = 0 \cdot -0 + 1(0 - (-1)) = 1$$

Ques Q_f is given. find V_L in terms of V_t (10)

tree branches = C, V_s & R_1

$$Q_f = \begin{bmatrix} & C & V_s & R_1 & R_2 & R_3 & L \\ \begin{matrix} 1 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 1 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 1 \end{matrix} & \begin{matrix} 0 \\ 0 \\ -1 \end{matrix} & \begin{matrix} 1 \\ 0 \\ 0 \end{matrix} & \begin{matrix} -1 \\ 0 \\ 1 \end{matrix} \end{bmatrix}$$

$$Q_f = [U : Q_{fL}]$$

$$Q_{fL} = \begin{bmatrix} & R_2 & R_3 & L \\ \begin{matrix} 0 \\ 1 \\ -1 \end{matrix} & \begin{matrix} 1 \\ 0 \\ 0 \end{matrix} & \begin{matrix} -1 \\ 0 \\ 1 \end{matrix} \end{bmatrix}$$

$$B_f = [-Q_{fL}^T : U]$$

$$B_f = \begin{bmatrix} & C & V_s & R_1 & R_2 & R_3 & L \\ \begin{matrix} 0 \\ -1 \\ 1 \end{matrix} & \begin{matrix} -1 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 1 \\ 0 \\ -1 \end{matrix} & \begin{matrix} -1 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 1 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 1 \end{matrix} \end{bmatrix}$$

$$B_f V_b = 0$$

$$B_f \begin{bmatrix} V_C \\ V_s \\ R_1 \\ R_2 \\ R_3 \\ L \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} & C & V_s & R_1 & R_2 & R_3 & L \\ \begin{matrix} 0 \\ -1 \\ 1 \end{matrix} & \begin{matrix} -1 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 1 \\ 0 \\ -1 \end{matrix} & \begin{matrix} -1 \\ 0 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 1 \\ 0 \end{matrix} & \begin{matrix} 0 \\ 0 \\ 1 \end{matrix} \end{bmatrix} \begin{bmatrix} V_C \\ V_s \\ V_{R_1} \\ V_{R_2} \\ V_{R_3} \\ V_L \end{bmatrix} = 0$$

$$-V_s + V_{R_1} + V_{R_2} = 0$$

$$V_{R_2} = V_s - V_{R_1} \quad - (1)$$

$$V_{R_3} = V_C \quad - (2)$$

$$V_L = -V_C + V_{R_1} \quad - (3)$$

(10)

Impedance of network N is obtained as

$$Z_{Th} = Z_{Nl} = \frac{V_{Th}}{I_N}$$

Case III : If circuit having only dependent sources:

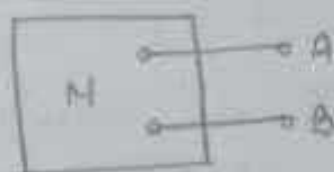
In this case, we apply a voltage V at terminal pair AB . A current I will flow due to application of V then the internal impedance of the network N is obtained

$$Z_{Th} = Z_{Nl} = \frac{V}{I}$$

v) Draw the equivalent circuit across AB

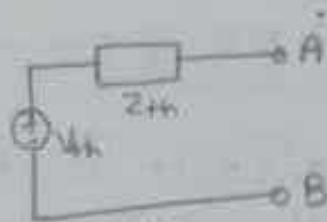
(a) Thevenin's voltage V_{Th} in series with Z_{Th}

(b) Norton's current I_N in parallel with Z_{Th} .



(a)

An active network N with terminal A & B



(b)

Thevenin's equivalent of network N



(c)

Norton's equivalent of network N .

Applications of Thevenin's and Norton's theorems:-

The one of the main application of Thevenin's and Norton's theorems is the replacement of a large part of a network, often a complicated and uninteresting part, by a very simple equivalent.

Limitations:- i) Not applicable to the circuits consisting of unilateral and non-linear elements like diode etc.

ii) Not applicable to the circuits consisting of load series or parallel with controlled or dependent sources.

Norton's theorem states that:

" w.r.t the terminal pair AB, the network N may be replaced with a current source I_N in parallel with an internal impedance Z_N . The current source I_N , called the Norton's current, is the current that would flow from A to B when the terminals A and B are shorted together and Z_N is the same internal impedance is defined earlier in context of Thevenin's theorem".

Procedure to obtain V_{th} and Z_{th} or I_N and Z_N :-

- i) Remove the portion of the network across which the Thevenin's or Norton's equivalent circuit is to be found.
- ii) Mark the terminals AB of the remaining 2-terminal circuit.
 - a) obtain the open circuit voltage at the terminal AB, i.e., V_{th} keeping all the sources at their normal values.
 - b) Short the terminals AB. obtain the short circuit current, i.e. I_N flowing from A to B keeping all the sources of their normal values.
- iii) Calculate $Z_{th} = Z_N$
 - Case I:- If circuit having only independent sources:-
In this case, set all the sources at zero values, i.e. with all the voltage sources shorted and all the current sources open circuited leaving behind their internal resistance or impedance.
 - Case II:- If circuit having independent and dependent source both:- In this case, we calculate V_{th} and I_N ; then the internal

two voltage in different parts of a circuit or between currents

Applications:- It is applicable to any linear, time invariant network composed of passive network elements. This theorem permits the interchange of the positions of the excitation and response. It provides the bilateral property of the network. It provides great convenience in design and measurement problems.

Limitations:-

- i) Not applicable to the network consisting of any dependent source even if it is linear.
- ii) Not applicable to the network consisting of any time varying elements.
- iii) Not applicable to the network consisting of non-linear elements like diode, transistor etc.

3] Thevenin's and Norton's Theorems:-

Thevenin's theorem states that:

"w.r.t. terminal pair AB; the network N may be replaced by a voltage V_{th} in series with an internal impedance Z_{th} . The voltage source V_{th} called the Thevenin's voltage is the potential difference ($V_A - V_B$) between the terminals A and B. and Z_{th} is the internal impedance of the network N as seen from the terminals A and B with all the sources set to zero, i.e. with all the voltage sources shorted and all the current sources open circuited leaving behind their internal resistances.

In Node Voltage Analysis, Kirchhoff's Current Law is used, which states that the algebraic sum of all incoming currents at a node must be equal to the algebraic sum of all outgoing currents at that node. It is the method of finding the potential difference between the elements or branches in a electric circuit.

→ This method has two types of nodes. - These are the non-reference node and the reference node.

The non reference nodes have a fixed voltage, and the reference node is the reference points for all other nodes.

→ In the nodal Method, the no of independent node pair equations needed is one less than the no of junctions in the network. That is if n denotes the number of independent node equations and j is the no of junctions.

$$n = j - 1$$

Network Theorems in AC Circuits :-

1) Super Position Theorem:- It states that: "In an active linear network containing several sources (including dependent source), the overall response (branch current or voltage) in any branch in the network equals the algebraic sum of the responses of each individual source considered separately with all other sources made inoperative, i.e replace by their internal resistances or impedances."

1) Mesh Current Analysis :- The Method that is used to determine the current flowing around a mesh or loop in the closed complex circuits is known as mesh analysis or mesh current analysis. It makes use of Kirchhoff's voltage law and Ohm's law to determine the unknown current flowing in a given complex circuit. For planar circuits ~~flowing in a given comp.~~ this analysis is very easy to use. We can find the ~~analysis~~ unknown voltage in a given circuit using Ohm's law.

Applications :-

- Solving of complex electrical circuit or networks to determine the unknown source like voltage and current with possible mesh equations.
- Simplifies the complex circuit.
- Reduces the mathematical equations with simplified calculations to solve the circuit.
- Analysis of complex circuits or networks.
- Determines the unknown current or mesh current, which flows through the meshes in the given circuit.
- Uses KVL and KCL to find the set of mesh equations to solve the circuit.

2) Node Voltage Analysis :- The Node Voltage Analysis is a method to solve the electrical network. It is used where it is essential to compute all branch currents. The node voltage analysis method determines the voltage and current by using the nodes of the circuit.

Applications of Super-position Theorem:- The Super position theorem is applicable

Applicable for any linear circuit having time varying or time invariant elements. It is useful in circuit analysis (i.e. finding the values of current and voltage in any branch of the circuit) when the circuit has a large number of independent sources, as it makes it possible to consider the effect of each source separately.

Limitations of Super position Theorem:-

- i) Not applicable to the circuits consisting of only dependent sources.
- ii) Not applicable to the circuit consisting of non-linear elements like diode, transistor etc.
- iii) Not applicable for calculation of power, since the Power is proportional to the square of current or voltage.

2) Reciprocity Theorem:- It is applicable to the linear networks only. It states that:

"The ratio of excitation to response remains invariant in a reciprocal network with respect to an interchange between the points of applications of excitations and measurement of response"

- Reciprocity theorem, simply indicates that voltage (V) and current (I) are mutually interchangeable. The ratio V/I is called the transfer resistance (or impedance). ~~As it~~ It is applied between a voltage and a current, either being the excitations and the other the response. It does not apply however, between